PHASE 5 — Critic and Evidence Verification

Objective

Implement a review mechanism that evaluates generated answers before they are returned to the user.

This is a pharmacology research application. Reliability and evidence grounding are critical.

Step 1 — Define critic responsibilities

The critic must examine:

- The original user question.
- The generated answer.
- The retrieved evidence.
- The source metadata.
- The citations associated with the answer.

It should identify:

- Unsupported claims.
- Claims that contradict the retrieved evidence.
- Missing citations.
- Invalid source references.
- Ambiguous clinical statements.
- Important omitted qualifications.
- Insufficient evidence.

Step 2 — Separate deterministic validation from LLM review

Do not rely exclusively on another LLM call to decide whether an answer is correct.

Implement deterministic checks for:

- Whether cited source identifiers exist.
- Whether citation references belong to the retrieved evidence.
- Whether citation metadata is valid.
- Whether required source information is present.

Use an LLM critic for semantic assessment of whether claims are supported by evidence.

Step 3 — Define review results

Use a structured result.

Possible statuses:

- APPROVED
- NEEDS_REVISION
- INSUFFICIENT_EVIDENCE

Include concise reasons and relevant claim/evidence references.

Step 4 — Revision and abstention

If the critic identifies issues:

- Allow a bounded revision attempt when sufficient evidence exists.
- Do not allow unlimited revision loops.
- If the evidence is insufficient, return an explicit limitation.
- Do not invent missing information to satisfy the question.

Step 5 — Clinical ambiguity

The application must handle ambiguous or incomplete clinical questions carefully.

It must not infer missing patient information.

It must distinguish between:

- General drug-label information.
- Patient-specific medical decisions.
- Information not available in the corpus.

The application must not present itself as a healthcare professional.

Step 6 — Tests

Test:

- Fully supported answers.
- Unsupported claims.
- Invalid citations.
- Contradictory evidence.
- Insufficient evidence.
- Ambiguous questions.
- Critic failures.
- Revision limits.

Mock LLM critic responses where appropriate.

Constraints

- Do not treat an LLM review as proof of medical correctness.
- Do not invent evidence.
- Do not remove citations during revision.
- Do not create unlimited review loops.
- Do not rewrite unrelated modules.

Deliverables

Explain:

1. The critic architecture.
2. Deterministic versus semantic validation.
3. How revisions work.
4. How abstention works.
5. Limitations.
6. Test results.